NeuroYou — Identity-Sensitive Model Examples

1 Case Introduction

This example is drawn from a two-year ethnographic study of an ML vendor, referred to as NeuroYou, developing an AI hiring system for a large international FMCG company, referred to as MultiCo [van den Broek et al. 2021]. NeuroYou's online games measured over 150 candidate attributes (personality traits, cognitive skills, career values); a supervised learning algorithm predicted candidate–organization fit by matching these attributes against profiles of high-performing employees. MultiCo processed over 10,000 trainee candidates per year.

In our models, Leo represents an ML developer/data scientist at NeuroYou whose professional identity is grounded in producing objective, data-driven knowledge independent of domain experts. Diana represents an HR manager at MultiCo whose professional identity is grounded in tacit hiring judgment, organizational culture knowledge, and the authority to define what constitutes a good employee. The Senior Manager position represents the final decision-makers who assessed candidates in group interviews. NeuroYou and MultiCo HR are modeled as positions covering Candidate Screening and Final Hiring respectively.

Unlike the Santander case (where identity threat was one-sided, affecting domain experts only) and the CleverLoan case (where two actors in the same profession responded differently to the same AI system), the NeuroYou setting introduces bilateral identity tension between two fundamentally different professional groups. The ML developers' identity is also at stake: their paradigm of producing knowledge independent of domain experts is repeatedly challenged by the need to include domain expertise. This cross-professional tension and the emergent hybrid outcome it produces are central to what the third case demonstrates.

2 Identity Tensions

The development and deployment of NeuroYou's ML system generated three successive problems, each triggering a cycle of mutual learning between developers and domain experts:

- **Data-selection problem** (Cycle 1: Constructing training data): The ML developers initially sought to collect all available employee data and let the algorithm determine which attributes mattered. MultiCo's HR team confronted them with the reality that without knowing which KPIs defined a "good employee," the developers would select irrelevant attributes. This forced the developers to include HR's professional standards—job descriptions, cultural ideals, and performance indicators—narrowing the initial 150 attributes to 49. For the developers, this compromised their independence; for the HR professionals, it required externalizing tacit knowledge that had never been formally articulated.

- **Frame problem** (Cycle 2: Building a model): The algorithm could only learn from the data developers supplied, producing a backward-looking model constrained by historical patterns. When the model revealed that cognitive skills did not predict performance and that valuing "good working conditions" correlated negatively with success, HR professionals objected: these findings conflicted with their vision of the future workforce. The developers had to incorporate

HR's forward-looking vision, adding non-predictive variables to align the model with organizational strategy. For the developers, this meant including subjective judgment they had sought to eliminate; for the HR professionals, it meant confronting evidence that their existing practices contained hidden biases (such as systematically rejecting introverted candidates who actually performed well).

- **Mismatch problem** (Cycle 3: Using the model in practice): Senior managers rejected many candidates the algorithm had predicted as exceptional fits. The developers realized that algorithmic predictions would not be acted upon without matching how managers reasoned. To bridge this gap, they introduced written feedback on candidates—changing a core aspect of MultiCo's hiring practice where decisions had previously been made through informal consensus. For the developers, this was a further departure from ML independence; for the managers, it required externalizing judgment that had always been tacit and relational.

This divergent, dialectical process—neither a designed collaboration (Santander) nor an unaddressed failure (CleverLoan)—produced a human–ML hybrid hiring practice through emergent mutual learning. Both groups' professional identities transformed: neither preserved intact, neither destroyed.

3 Functional 'As-Is' SR Model

What this model conveys:

- The ML-assisted hiring process is modeled using agents (ML Developer, Hiring Expert, HR Professional, Job Candidate) and positions (NeuroYou, MultiCo HR, Senior Manager), without person-based identities.
- The ML Developer agent OCCUPIES NeuroYou (position), which COVERS Candidate Screening. The Hiring Expert agent PLAYS into the Candidate Screening boundary, which contains the full screening workflow: Collect Game Responses, Select Predictive Attributes, Generate Candidate-Fit Predictions, and Provide Prediction to Final Hiring.
- Select Predictive Attributes is the identity-critical task—the primary locus where the data-selection problem will emerge. In this functional view, it appears purely operational.
- The HR Professional agent PLAYS the Final Hiring role and OCCUPIES the Senior Manager position, who is PART of MultiCo HR (position) that also COVERS Final Hiring. Final Hiring contains the decision workflow: Review AI Predictions, Conduct Group Interview, and Make Hiring Decision.
- Softgoals at the role level—Objective Assessment, Efficient Screening, Accurate Prediction for Candidate Screening; Traceability and Efficiency for Final Hiring—represent functional quality concerns only.
- Job Candidate is an external agent connected through dependencies: Candidacy be Assessed (goal) and Game Responses (resource).

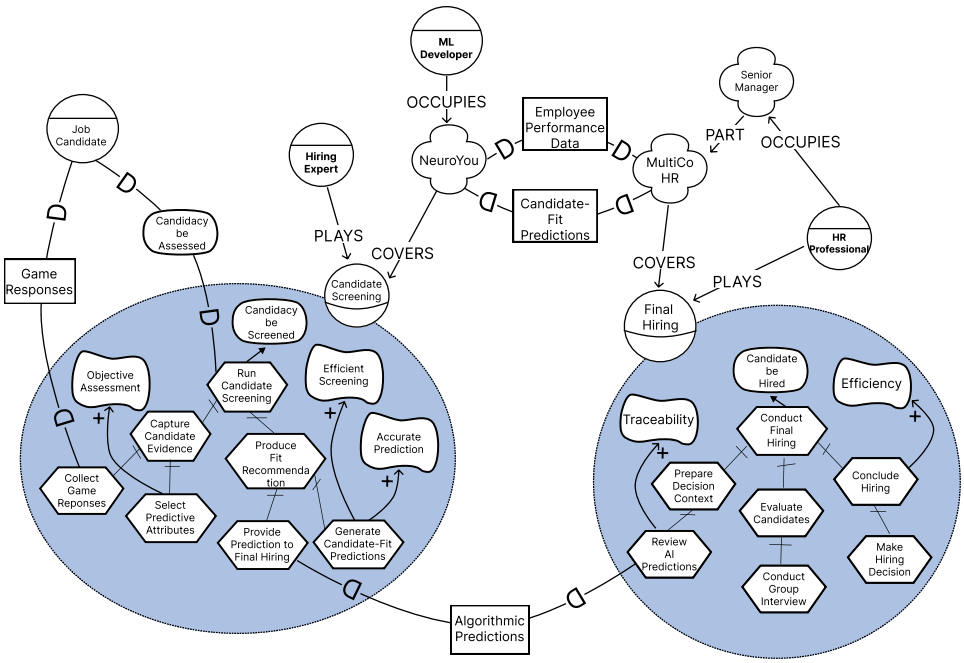

Figure 1: NY-1: Functional 'As-Is' SR Model.

Key analytical point. A purely functional analysis sees the hiring pipeline as a data flow from game responses through algorithmic prediction to managerial decision. There is no functional reason not to maximize ML independence—more data, less expert input, better predictions. The model does not capture that Select Predictive Attributes requires tacit domain knowledge that HR professionals possess but cannot easily formalize, that the ML developers and HR professionals hold fundamentally different epistemologies about what constitutes valid knowledge, or that senior managers' interview judgments may systematically diverge from algorithmic predictions.

4 Identity-Sensitive 'As-Is' SR Model

What this model conveys:

- Two person-based agents are introduced: Leo (ML developer/data scientist at NeuroYou, upper left) and Diana (HR manager at MultiCo, upper right), each instantiating their respective role with fundamentally different professional identities.
- Leo's boundary contains a goal (ML Model be Trained) and a main task (Build Screening Model) that decomposes into two either/or alternatives. Alt 1: Independent Build (Keep Modeling Logic Internal, Train and Score Independently) preserves his data-driven independence. Alt 2: Collaborative Build (Align Features with HR, Refine with HR Feedback) addresses relevance but compromises independence. His identity-sustaining softgoals—Scientific Objectivity (+), Independence from Domain Bias (+), Data-Driven Innovation (+), Algorithmic Superiority (+)—reflect the ML paradigm's core self-concept.
- Diana's boundary contains a goal (Hiring Criteria to be Set) that decomposes into two either/or alternatives. Alt 1: Tacit Criteria (Keep Criteria Implicit, Prioritize Human Judgment) preserves her tacit expertise monopoly. Alt 2: Explicit Criteria (Codify Standards for Model Use, Revise Criteria from Model Feedback) enables ML relevance but requires externalizing tacit knowledge. Her identity-sustaining softgoals—Professional Judgment (+), Domain Authority (+), Organizational Vision (+), Hiring Expertise (+), Hiring Efficiency (−/+)—reflect established HR expertise.

- The role-level boundaries (Candidate Screening, Final Hiring) and position-level structure (NeuroYou, MultiCo HR) are preserved from NY-1—what changes is the identity context overlaid through person-based agents.
- Two reciprocal dependencies connect Leo and Diana: Leo depends on Diana for HR Standards (resource), and Diana depends on Leo for Algorithmic Insights (resource). Under Alt 1, both dependencies are dormant; under Alt 2, both activate. This reciprocal pair is the structural signature of mutual learning.

Key analytical point. The bilateral identity tension is the distinctive feature. Leo's Alt 1 (independence) fails because he hits the data-selection, frame, and mismatch problems—each requiring domain expertise to resolve. Diana's Alt 1 (traditional practice) fails because MultiCo needs the efficiency and scale that only ML can provide. Neither actor can succeed independently, but collaboration requires each to compromise their professional self-concept. The reciprocal dependency pair makes this tension structurally explicit.

What this reveals beyond Santander and CleverLoan. The Santander As-Is model had a single protagonist (Tim) whose identity was threatened by AI. The CleverLoan As-Is model showed two actors within the same profession (experienced vs. junior consultant) with divergent baseline identities. The NeuroYou As-Is model shows two actors from different professions whose professional identities are epistemologically opposed—and crucially, the AI builders' identity is also at stake, a dynamic not modeled in either previous case. This tests the framework's ability to represent cross-professional identity tension and bilateral identity threat.

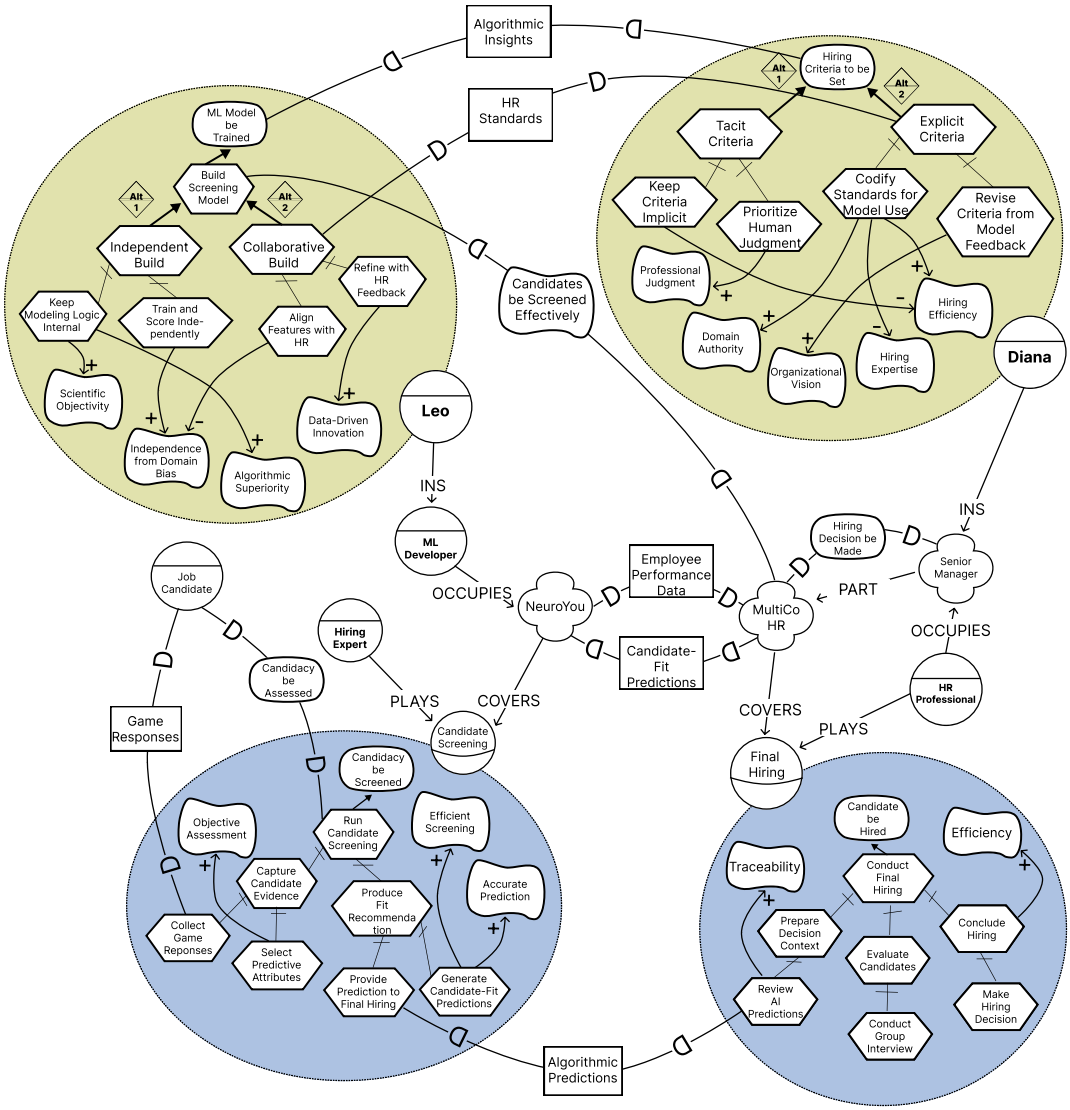

Figure 2: NY-2: Identity-Sensitive 'As-Is' SR Model.

5 'To-Be' SD Model with ALT 1 and 2

What this model conveys:

- Two ISA splits restructure As-Is agents into To-Be specialized agents. On the left: ML Developer (from NY-1) ISA → ML Screening Specialist—the algorithmic screening function. On the right: Hiring Expert (from NY-1) ISA → Domain Knowledge Expert—the human expertise function. Additionally, HR Professional ISA → Hiring Expert shows the generalization chain on the right side. Both ISA bases come from the As-Is model, so the reader can trace exactly what was restructured and why. This parallels the Santander and CleverLoan ISA splits—in all three cases, AI introduction decomposes existing professional agents into specialized human and AI variants.
- Person-based agents are preserved: Leo and NeuroYou ML System both INS → ML Developer; Diana INS → Domain Knowledge Expert.

- ML Screening Specialist OCCUPIES NeuroYou (position), which COVERS Candidate Screening. HR Professional OCCUPIES Senior Manager (position), which is PART of MultiCo HR (position); MultiCo HR COVERS both Candidate Screening and Final Hiring.
- Shared dependencies (black): Job Candidate provides Game Responses to Candidate Screening; MultiCo HR depends on Senior Manager for Hiring Decision be Made; and Job Candidate depends on Candidate Screening for Fair Assessment.
- Alternative 1: ML Independence (blue) shows a single dependency—Senior Manager receives Raw Prediction Scores from ML Screening Specialist. Domain Knowledge Expert has no dependency connections. Diana is structurally excluded from the screening process.
- Alternative 2: Hybrid Practice (red) shows five dependencies. B1: ML Screening Specialist depends on Domain Knowledge Expert for HR Standards. B2: Domain Knowledge Expert depends on ML Screening Specialist for Algorithmic Insights. B3: Senior Manager depends on ML Screening Specialist for Contextualized Predictions. B4: ML Screening Specialist depends on Senior Manager for Written Manager Feedback. B5: Domain Knowledge Expert depends on NeuroYou for Employee Performance Data—the training data dependency, which under the hybrid practice flows through the Domain Knowledge Expert rather than directly to the ML system.
- Domain Knowledge Expert carries the softgoal Domain Authority be Recognized—visible only under Alt 2.
- B1 and B2 form the reciprocal pair—the SD-level expression of mutual learning. Under Alt 1, one blue line; under Alt 2, five red lines. The visual contrast between the sparse Alt 1 and the densely connected Alt 2 makes the structural consequence of mutual learning immediately legible.

Key analytical point. Unlike Santander (where the TA manager deliberately chose the collaborative alternative) and CleverLoan (where the Branch Manager defaulted toward full AI delegation), the NeuroYou case shows that Alt 1 fails in practice. The ethnography documents three successive failures of ML independence, each requiring progressively deeper inclusion of domain expertise. Alt 2 is not a pre-designed governance choice but an emergent outcome—the dialectical result of repeated confrontation with the technology.

What this reveals beyond Santander and CleverLoan. The Santander To-Be SD split one agent into Human/AI variants. The CleverLoan To-Be SD did this and created an entirely new agent (AI-Supported Consultant). The NeuroYou To-Be SD uses two parallel ISA splits—ML Developer ISA $\rightarrow$ ML Screening Specialist and Hiring Expert ISA $\rightarrow$ Domain Knowledge Expert—both originating from As-Is agents. This introduces a structural asymmetry: Domain Knowledge Expert exists in both alternatives but is connected through dependencies only under Alt 2. The presence or absence of its dependency lines is the SD-level signature of whether mutual learning occurs—testing the framework's ability to represent conditional participation as a design choice.

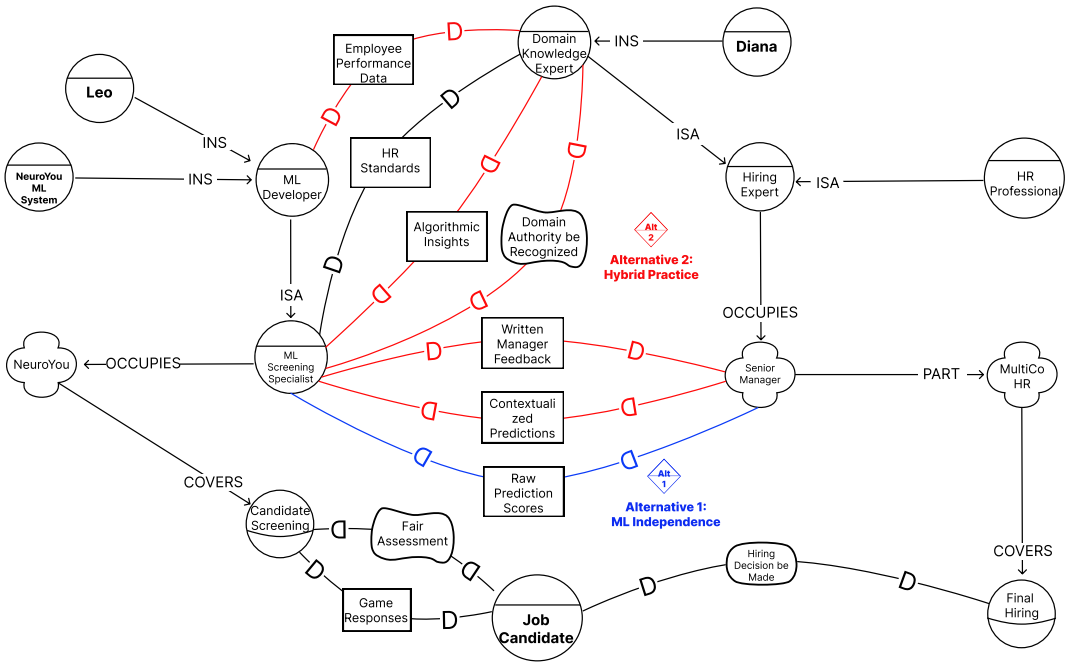

Figure 3: NY-3: 'To-Be' SD Model with ALT 1 and ALT 2.

6 'To-Be' SR Model with ALT 2

What this model conveys:

- The NY-4 layout inherits the NY-1 skeleton: Candidate Screening boundary (blue, left) with the full screening task decomposition, and Final Hiring boundary (blue, right) with the decision workflow. Positions (NeuroYou, MultiCo HR, Senior Manager), agents (ML Developer, Hiring Expert, HR Professional), and structural links are all preserved. Leo connects through the left ISA chain: Leo INS → ML Developer ISA → ML Screening Specialist. Diana connects through the right ISA chain: Diana INS → Domain Knowledge Expert ISA ← Hiring Expert ISA ← HR Professional.

- Leo's boundary (green, upper left) shows the hybrid compromise. His goal is Hybrid Model be Operational and his main task Build Domain-Engaged Model decomposes into four tasks: Consult HR on Attribute Selection, Incorporate Expert Standards into Training Data, Present Model Findings to HR Team, and Revise Model Based on Expert Feedback. His identity-sustaining softgoals show partial satisfaction: Scientific Objectivity ($\sim$), Data-Driven Innovation ($+$), Algorithmic Relevance ($+$), Independence from Domain Bias ($--$), and the emergent relational softgoal Mutual Understanding ($+$).

- Diana's boundary (green, upper right) shows knowledge externalization and reflective growth. Her goal is Hiring Criteria be Co-Developed and her main task Externalize and Evolve HR Expertise decomposes into four tasks: Articulate Professional Standards for ML, Provide Performance Indicators, Reflect on Algorithmic Insights, and Advocate for Future Workforce Vision. Her identity-sustaining softgoals show: Professional Judgment ($\sim$), Domain Authority ($+$), Bias Awareness ($++$), Hiring Expertise ($-$), and the emergent relational softgoal Reflective Practice ($+$).

- The Senior Manager's boundary adds a new task not present in NY-1: Provide Written Feedback on Candidates, which externalizes reasoning that was previously informal and feeds the reciprocal learning loop.
- The Alt 2 dependencies from NY-3 (B1–B5) are expressed as resource flows between the SR boundaries: HR Standards flows from Diana to Leo; Algorithmic Insights flows from Leo to Diana; Contextualized Predictions flows from Leo to Senior Manager; Written Manager Feedback flows from Senior Manager to Leo; Employee Performance Data flows from NeuroYou through Diana (the Domain Knowledge Expert mediates what training data is relevant).
- The strongest negative contribution in the model is Independence from Domain Bias (−−), reflecting that the hybrid practice fundamentally compromises the ML independence paradigm. The strongest positive contribution is Bias Awareness (++), reflecting that algorithmic insights trigger the most productive identity transformation: Diana discovers that her team had systematically rejected introverted candidates who actually performed well, that cognitive skills were not predictive of success, and that managers suffered from anchoring bias.

Key analytical point. The To-Be SR with Alt 2 models identity transformation—neither preservation (Santander) nor destruction (CleverLoan), but dialectical evolution through emergent mutual learning. Both Leo and Diana's professional self-concepts change in ways neither anticipated. Leo gains a relational dimension (Mutual Understanding) that goes beyond his original independent-scientist identity. Diana gains a reflective dimension (Reflective Practice, Bias Awareness) that enriches her practice beyond her original tacit-expert identity. The (−−) on Independence from Domain Bias and the (++) on Bias Awareness are the structural signatures of this third trajectory.

What this reveals beyond Santander and CleverLoan. The Santander To-Be SR showed how identity-sensitive design choices resolve tensions through transparency, authority preservation, and competence development—all positive outcomes achieved through deliberate design. The CleverLoan To-Be SR showed the costs of ignoring identity: productive adaptation and destructive resistance coexisting in the same actor, scope expansion without decision authority, and organizational behaviors invisible to functional requirements. The NeuroYou To-Be SR shows a third pattern: identity transformation where both the strongest negative contribution (−−) and the strongest positive contribution (++) appear in the same model, reflecting that genuine mutual learning requires both loss and growth. No previous case produced emergent relational softgoals (Mutual Understanding, Reflective Practice) that have no precedent in the As-Is—these are identity elements that the hybrid practice itself creates.

7 Coverage Analysis: What NeuroYou Adds to the Three-Case Arc

Following the software testing analogy, the table below tracks which aspects of the identity-sensitive modeling framework are exercised by each case:

This completes the three-case arc. Santander shows that identity-sensitive design works when deliberately chosen. CleverLoan shows what happens when identity is ignored. NeuroYou shows a third path: identity transformation through emergent mutual learning, where neither pure automation nor pure domain control succeeds, and both professional groups must evolve.

Modeling capability	**Santander**	**CleverLoan**	**NeuroYou**	**Notes**
Person-based agent with identity softgoals	Tim (single protagonist)	Max + Anna (same profession, different experience)	Leo + Diana (different professions, opposed epistemologies)	Progressive complexity: 1 agent, 2 agents same role, 2 agents cross-profession
AI builder's identity at stake	Not tested	Not tested	Leo's independence paradigm challenged	Only NeuroYou tests this
Bilateral identity threat	Not tested (one-sided)	Not tested (divergent but not bilateral)	Both Leo and Diana face identity compromise	The reciprocal pair is the structural expression
Emergent relational softgoals	Not tested	Not tested	Mutual Understanding (Leo), Reflective Practice (Diana)	Identity elements created by the hybrid practice itself
Alt-dependent agent participation	Not tested	Not tested	Domain Knowledge Expert connected only under Alt 2	Tests conditional structural activation
Cross-organizational dependency	Not tested (internal)	Not tested (internal)	NeuroYou (vendor) $\leftrightarrow$ MultiCo HR (client)	Adds inter-organizational boundary
Identity trajectory	Designed preservation	Unaddressed destruction	Emergent transformation	Three-case arc: success $\rightarrow$ failure $\rightarrow$ negotiation

Table 1: Coverage analysis: what each case exercises in the identity-sensitive modeling framework.